

- 1 A rod is fixed horizontally. A solid bar of mass M is attached to the rod by two vertical strings so that the bar is horizontal, as shown in Figure 1.1.

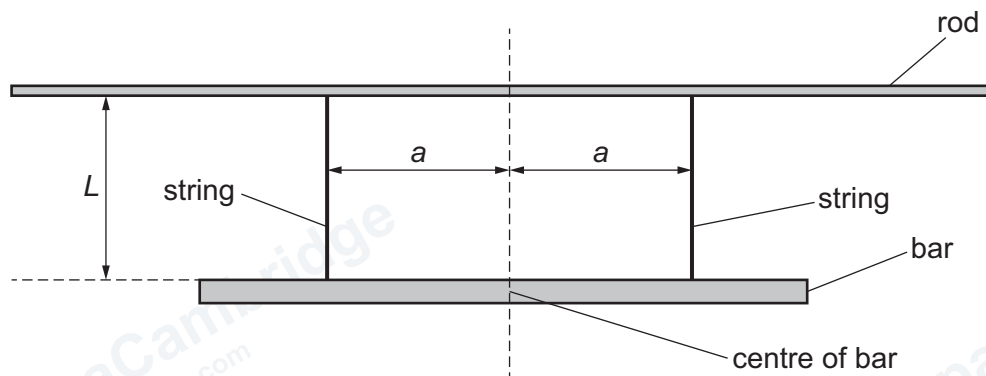


Figure 1.1

The vertical strings are equal horizontal distances a from the centre of the bar.

The separation between the rod and the bar is L .

One end of the bar is displaced horizontally and then released. The bar rotates with oscillations about its centre. The period of the oscillations is T .

It is suggested that T is related to a by the relationship

$$T = \frac{2\pi P}{a^q} \sqrt{\frac{L}{M}}$$

where P and q are constants.

Plan a laboratory experiment to test the relationship between T and a .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for P and q .

In your plan, you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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